components, showing how to arrange them in the correct way for your application. In chapter 31 you'll learn how to build your own middleware components and incorporate them into the pipeline.

We'll begin by looking at the concept of middleware, all the things you can achieve with it, and how a middleware component often maps to a cross-cutting concern. These functions of an application cut across multiple different layers. Logging, error handling, and security are classic cross-cutting concerns that are required by many parts of your application. Because all requests pass through the middleware pipeline, it's the preferred location to configure and handle this functionality.

In section 4.2 I'll explain how you can compose individual middleware components into a pipeline. You'll start out small, with a web app that displays only a holding page. From there, you'll learn how to build a simple static-file server that returns requested files from a folder on disk.

Next, you'll move on to a more complex pipeline containing multiple middleware. In this example you'll explore the importance of ordering in the middleware pipeline, and you'll see how requests are handled when your pipeline contains multiple middleware.

In section 4.3 you'll learn how you can use middleware to deal with an important aspect of any application: error handling. Errors are a fact of life for all applications, so it's important that you account for them when building your app.

You can handle errors in a few ways. Errors are among the classic cross-cutting concerns, and middleware is well placed to provide the required functionality. In section 4.3 I'll show how you can handle exceptions with middleware provided by Microsoft. In particular, you'll learn about two different components: